

Getting Started with Tabular Foundation Models with TabPFN

Thinking Mode, Interpretability & More

- PyLadies Amsterdam



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20 AUGUST 2026

01

What are Tabular Foundation Models?

Foundation Models have transformed text & images

But your most valuable data is organized in tables

Text



Databases
Spreadsheets

Csv
Time series

AGE	TENURE	REGION	SPEND	TARGET
41	18	EU	2 140	0
27	3	US	890	1
54	31	EU	4 305	0
33	7	APAC	1 620	?
62	12	US	3 070	?

PRIOR

Images



What is tabular prediction? A concrete example

The decade old problem formulation in computer science

City Code	Region Code	Accomodation Type	Upper Age	Health Indicator	Reco Policy Category	Response
C8	2190	Rented	44	X2	16	0
C9	1785	Rented	52	X2	22	1
C3	679	Owned	28		17	0
C1	3175	Owned	75	X4	17	1
C15	3497	Owned	52	X1	1	0
C1	530	Owned	59		18	1
C28	600	Owned	21	X2	21	1
C27	1097	Owned	59	X3	13	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮
C3	1484	Rented	20	X3	2	0
C24	2748	Owned	67	X1	17	?
C27	811	Owned	75	X1	20	?
C2	256	Owned	42	X1	20	?

X_{train}

y_{train}

X_{test}

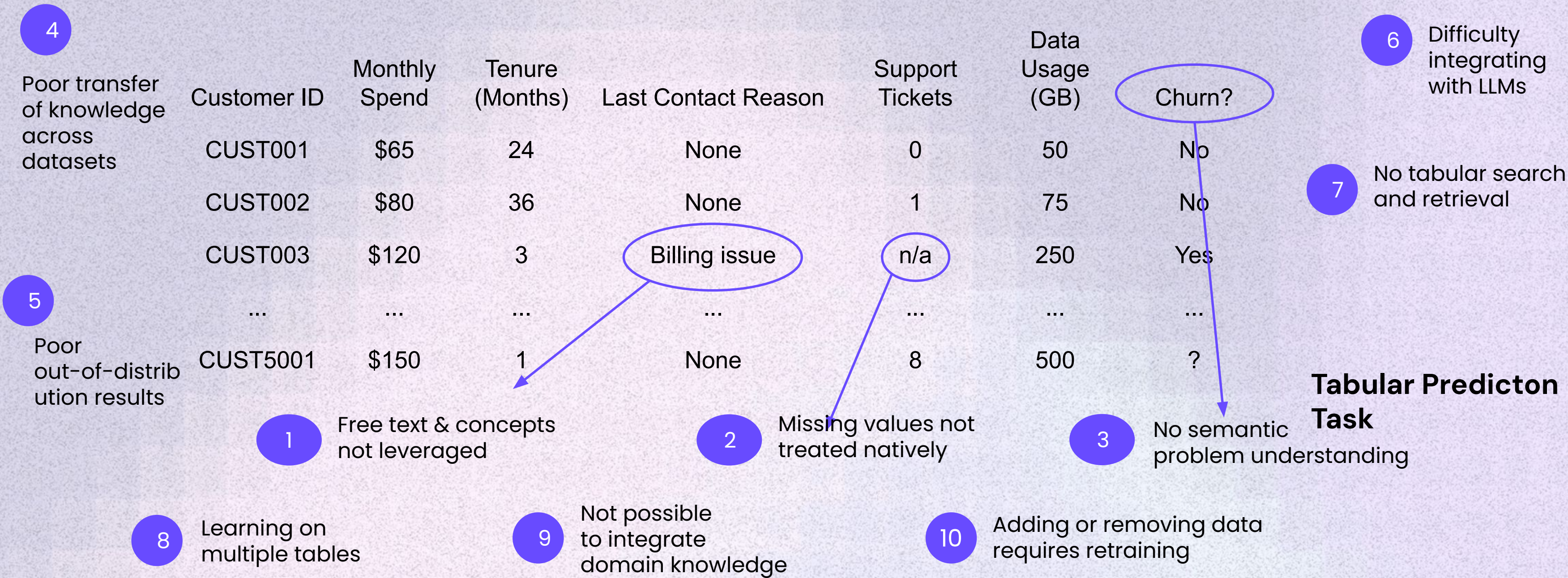
y_{test}

Other examples (among 1000s)

- Will this customer churn?
- Will this applicant repay their loan on time?
- Which product will this customer buy?
- ...

Tabular prediction is stuck in the 2000s

The decade old problem formulation in computer science



TabPFN was pre-trained on hundreds of millions of synthetic datasets

So your table is input to a model that already exists, not training material for a new one

ONCE, AT PRIOR LABS

0.41	B	1
0.08	A	0
0.77	B	1

100M+

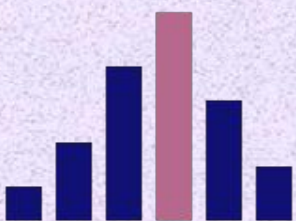
synthetic datasets,
weights update

The model learns how to learn from a table, not how to solve one table.

EVERY TIME, ON YOUR DATA

AGE	SPEND	CHURN
41	2 140	0
27	890	1
33	1 620	?

TabPFN

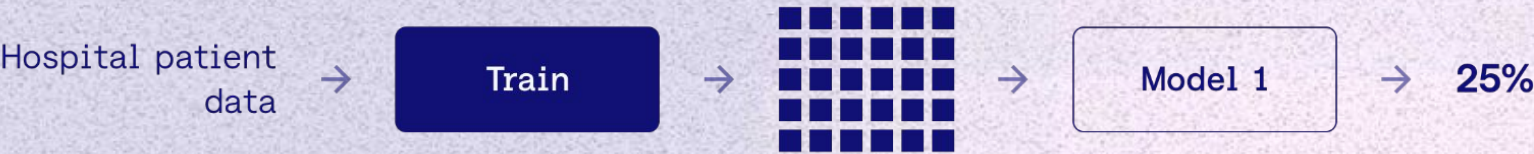


One forward pass, no gradient step, no tuning. Seconds to a prediction.

What If One Model Could Handle Every Prediction?

The foundation model revolution for tabular data

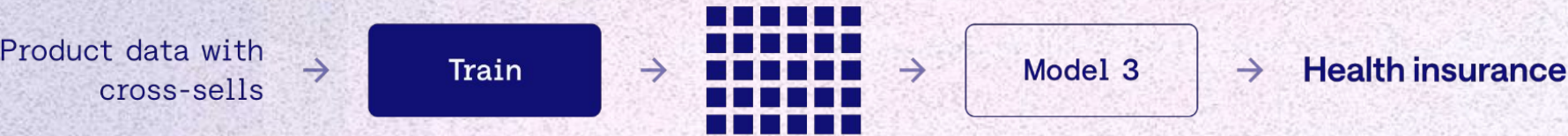
What is patient X's risk of deterioration?



What's that transaction Fred made?



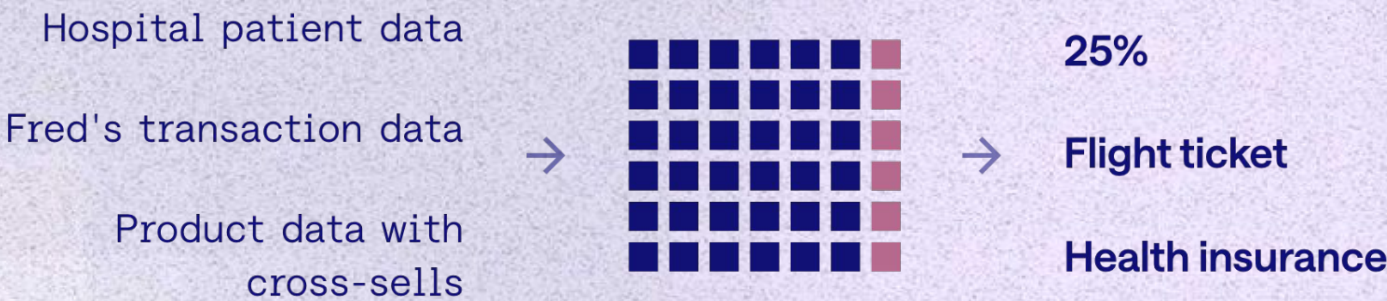
Which product does customer Y buy next?



ONE MODEL

TabPFN

Pre-trained foundation model. No training run, no tuning.



Recap

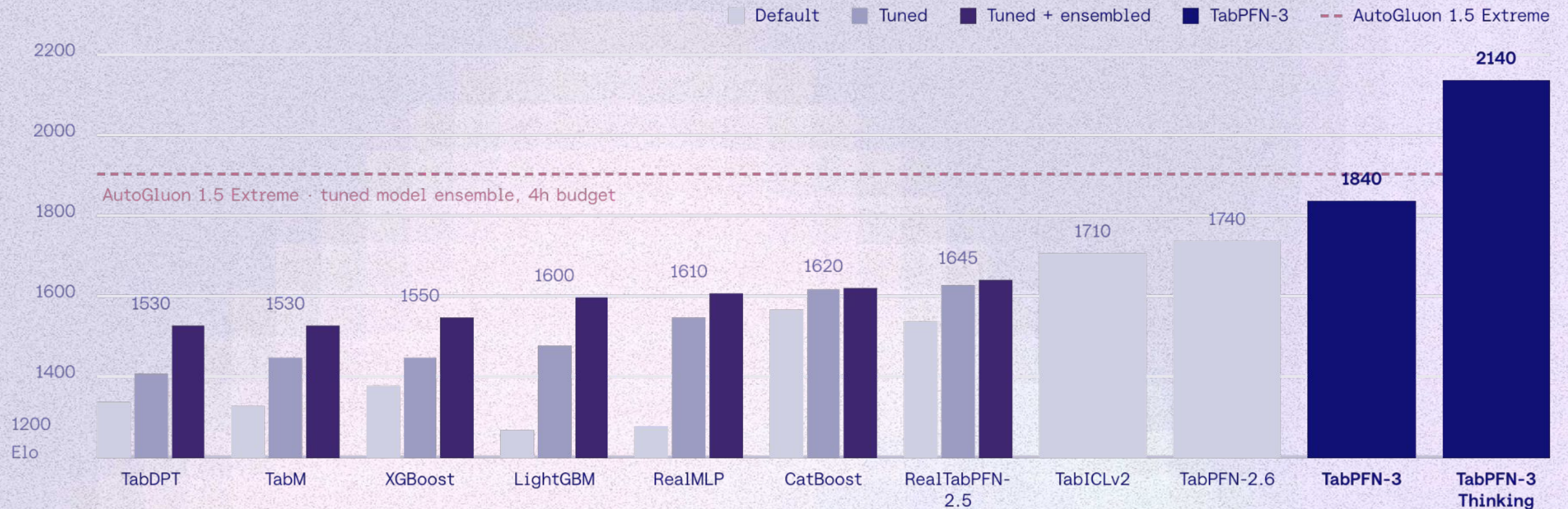
Four years, from research to production

Each release moved the same model further into real data: more rows, more columns, less tuning

JULY 2023 TabPFN v1 A transformer that solves small tabular problems in a second. Pre-print, then ICLR 2023. 1k rows, numeric features	DEC 2024 Prior Labs founded Prior Labs founded to develop the best-in-class TFMs to solve tabular prediction tasks Research to product	JAN 2025 TabPFNV2 in Nature Published in Nature. Categorical features, missing values and outliers: real data, not clean data. 10k rows, 500 features	NOV 2025 TabPFN-2.5 Leading method on TabArena, plus a distillation engine for low-latency deployment. 50k rows, 2k features	MAY 2026 TabPFN-3 Enterprise scale on the same training-free workflow. The current release. 1M rows, 200 features 1k rows, 20k features
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What If One Model Could Handle Every Prediction?

The foundation model revolution for tabular data

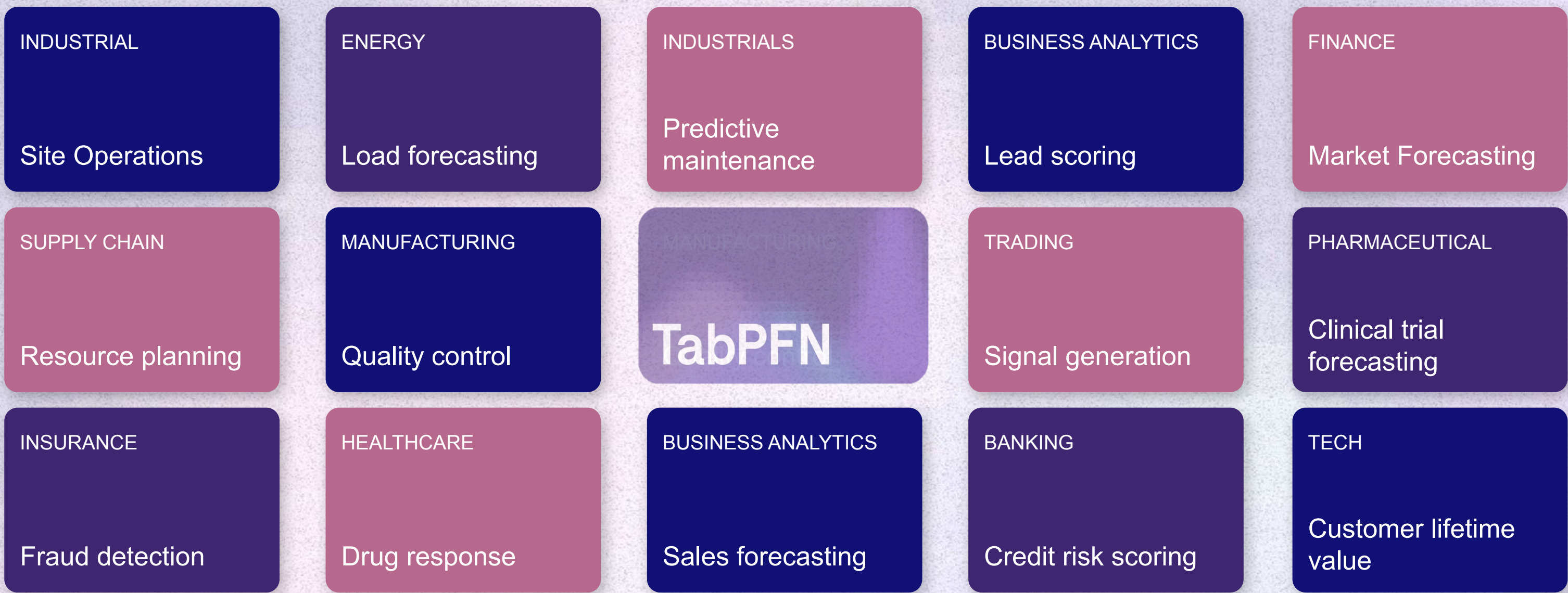


Performance on the TabArena benchmark [1], largest data subset (10k-100k samples). TabPFN-3 outperforms any other model in a forward pass. TabPFN-3-Plus (Thinking) is dramatically better yet, outperforming AutoGluon 1.5 extreme [2], a complex ensemble of models tuned for 4 hours, while being 10x faster.

Recap

TabPFN applies to all types of prediction tasks

One model to unlock tabular data across your business



02

Workshop